

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			award (1) for any of following - H₂SO₄ loses protons / H⁺ ions and NH₃ gains protons / H⁺ ions - H₂SO₄ is a proton donor and NH₃ is a proton acceptor - NH₃ accepts a proton from H₂SO₄ neutral answer – reference to salt formation			1	1		
	(b)			moles (NH ₄) ₂ SO ₄ = $\frac{1.86}{132.18}$ = 0.0141 (1) moles NaOH = 0.0282 (1) concentration = $\frac{0.0282}{0.0267}$ = 1.06 mol dm ⁻³ (1)		3		3	1 1	
	(c)	(i)		if a system at equilibrium is subjected to a change then the position of equilibrium will shift to minimise/oppose that change	1			1		
		(ii)		student correct since more moles of ammonia present at equilibrium at higher temperature (1) system opposes increase in temperature by shifting equilibrium position to the left / in the endothermic direction (1) neutral answer – reference to the ‘endothermic <u>side</u> ’ if incorrect award no marks accept converse answer student correct since fewer moles of ammonia present at equilibrium at lower temperature (1) system opposes decrease in temperature by shifting equilibrium position to the right / in the exothermic direction (1)	1		1	2		